

Exercise 6

1. What is (unsupervised) clustering? What is the difference between classification and clustering? Give examples from real life.
2. Describe the K-Means algorithm and its objective function. What is a cluster centroid? How is its position determined? How/When does the algorithm terminate? List the advantages and disadvantages of the algorithm.
3. The similarity/dissimilarity between objects can be described in different ways. Explain Manhattan distance, Euclidean distance and Cosine similarity. What are the formulas, value ranges and significance of these similarity functions?
4. Execute K-Means with $K = 2$ and Euclidean distance on the data below: assign all objects to a cluster, compute the SSE and find the centroid positions in each iteration. Instead of a random initialisation the initial centroids are p_{10} and p_{11} .

ID	x	y
p1	1.0	1.0
p2	2.0	2.0
p3	3.0	1.0
p4	3.0	2.5
p5	2.0	8.0
p6	2.0	6.0
p7	2.5	7.0
p8	4.0	7.0
p9	6.0	2.0
p10	7.0	3.0
p11	6.5	1.0

5. Explain the Bisecting K-means algorithm. How is it different from standard k-means? Explain how the process is carried out.